

2026
MATH CP
PAPER Paper 1

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Candidate Number

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HKDSE Mathematics Compulsory Part — Paper 1 (Mock)
PAPER Paper 1
Question-Answer Book

135 minutes (2.25 hours)

Total marks : 105

INSTRUCTIONS

This paper consists of THREE sections, A(1), A(2) and B. Unless otherwise specified, all working must be clearly shown. Unless otherwise specified, numerical answers should be either exact or correct to 3 significant figures. The diagrams in this paper are not necessarily drawn to scale.

Section A(1) (35 marks)

1. Simplify $\frac{p^7q^{-2}}{(p^{-1}q)^3}$ and express your answer with positive indices.

(3 marks)

2. Make h the subject of the formula $\frac{3h+2}{2h-5} = k$.

(3 marks)

3. (a) A packet of sugar is termed acceptable if its weight is measured as 340 g correct to the nearest 20 g . Find the least possible weight of an acceptable packet of sugar.
- (b) There are 5 acceptable packets of sugar. Is it possible that the total weight of these 5 packets is 1740 g ? Explain your answer.

(3 marks)

4.
 - ((a)) Factorize $16x^2 - 24xy + 9y^2$.
 - ((b)) Factorize $16x^2 - 24xy + 9y^2 + 8x - 6y$.

(4 marks)

5. An iron is sold at a discount of 20% on its marked price. After selling the iron, the profit is \$240 and the percentage profit is 30% . Find the marked price of the iron.

(4 marks)

6. ((a)) Consider the compound inequality $-2(x + 1) \geq x - 11$; **and** $4x + 1 > x - 5$; $\dots$; (*) . Solve (*) .
- ((b)) Write down the greatest integer satisfying (*) .

(4 marks)

- (4 marks)

8.

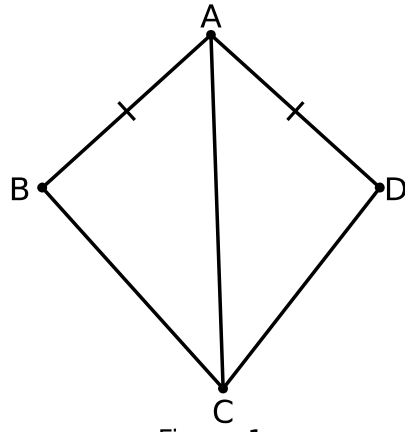


Figure 1

- (a) In Figure 1, ABCD is a quadrilateral. It is given that $AB = AD$ and $\angle BAC = \angle DAC$. Prove that $\triangle ABC \cong \triangle ADC$.
- (b) If $\angle ABC = 104^\circ$ and $\angle BCD = 88^\circ$, find $\angle BAD$.

(5 marks)

- ((a)) The table below shows the distribution of the times taken by a group of 40 workers to finish a product.

Time taken (minutes)	11-15	16-20	21-25	26-30	31-35
Frequency	5	11	13	7	4

Write down the modal class of the distribution.

- (b) Find the mean of the distribution.
- (c) A worker is randomly selected from the group. Find the probability that the time taken by the selected worker is less than 20.5 minutes.

(5 marks)

Section A(2) (35 marks)

10. ((a)) It is given that $f(x)$ is partly constant and partly varies as x^2 . Suppose that $f(1) = 9$ and $f(4) = 54$. Find $f(x)$.
- ((b)) Write down the y-intercept of the graph of $y = f(x)$.
- ((c)) Let k be a real constant. Find the range of values of k such that the equation $f(x) = k$ has two distinct real roots.

(7 marks)

Stem (tens)	Leaf (units)
1	3 6 9
2	2 5 a
3	1 4 b
4	0 3

((a)) The stem-and-leaf diagram in Figure 2 shows the distribution of the ages of the 11 members of a club. It is given that the median and the inter-quartile range of the distribution are 27 and 16 respectively. Find a and b .

((b)) Find the mean and the standard deviation of the distribution.

((c)) A member aged 43 now leaves the club. Find the change in the median of the distribution.

- (7 marks)

- (7 marks)

14. ((a)) The equation of the circle C is $x^2 + y^2 - 20x - 16y + 139 = 0$. Denote the centre of C by G . Find the coordinates of G and the radius of C .
- ((b)) The coordinates of the point H are (22, 8) . Find the distance between G and H .
- ((c) Let P be a moving point on C . When the area of $\triangle GHP$ is the greatest, describe the geometric
- (i) relationship between GH and GP .
- ((c) Find the perimeter of $\triangle GHP$ in this case.
- (ii)

(7 marks)

Section B (35 marks)

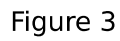
15. (a) In a box, there are 8 red balls, 4 green balls and 3 yellow balls. 4 balls are randomly drawn from the box at the same time. Find the probability that exactly 2 red balls are drawn.
- (b) Find the probability that at least 1 red ball and at least 1 yellow ball are drawn.

16. ((a)) Let $f(x) = 2x^2 - 8kx + 10k^2 + 8k + 5$, where k is a real constant. Using the method of completing the square, express, in terms of k , the coordinates of the vertex of the graph of $y = f(x)$.
- ((b)) Find the least value of the y -coordinate of the vertex and the corresponding value of k .
- ((c)) Someone claims that for any real constant k , the vertex of the graph of $y = f(x)$ lies on the curve $\Gamma : y = \frac{1}{2}x^2 + 4x + 5$. Do you agree? Explain your answer.

(6 marks)

17. ((a)) Let $A(n)$ be the n th term of an arithmetic sequence. It is given that $A(3) = 19$ and $A(8) = 44$. Find $A(n)$.
- ((b)) Find $A(1) + A(2) + \dots + A(45)$.
- ((c)) Let $G(n)$ be the n th term of a geometric sequence with first term $A(2)$ and common ratio 2. Find the least value of n such that $G(1) + G(2) + \dots + G(n) > A(1) + A(2) + \dots + A(45)$.

(6 marks)



- ((a)) Figure 3 shows a right pyramid VABCD with a square base ABCD of side 10 cm . The height of the pyramid is 12 cm . Let M be the mid-point of BC and O be the centre of ABCD . Find VM .
- ((b)) Find the angle between the plane VBC and the base ABCD .
- ((c)) Let N be the mid-point of AD . A student claims that the angle between the plane VBC and the plane VAD is greater than 45° . Do you agree? Explain your answer.

(8 marks)

19. ((a)) The equation of the circle C is $x^2 + y^2 - 6x - 8y = 0$. Denote the centre of C by G. Find the coordinates of G and the radius of C.
- ((b)) Let $L : 3x + 4y - 50 = 0$. Prove that L is a tangent to C, and find the coordinates of the point of contact T.
- ((c)) Denote the origin by O. Someone claims that OT is a diameter of C. Do you agree? Explain your answer.
- ((d)) Let P be a moving point on C such that P is different from O and T. Denote the area of $\triangle OPT$ by S. Find the greatest value of S.

(9 marks)

END OF PAPER